MATH3303 – Lecture 13

Classification of Finite Abelian Groups IV: Lemma 7 and the Fundamental Theorem

Review Notes

Reference: Lee, Abstract Algebra, *§5.2*

Overview

- **Lemma 7** (Lee, Lemma 5.6): complete proof that a finite abelian p-group splits as $G = \langle g \rangle \times H$ where g has largest order. This is the hardest technical result in the classification, presented in full with four clearly labelled steps.
- **Fundamental Theorem of Finite Abelian Groups** (Lee, Thm. 5.3): statement. The proof follows in Lecture 14.

1 Lemma 7: Splitting off the Largest-Order Element

Lemma 1.1: Splitting off largest-order element (Lemma 7 from lecture; cf. Lee, Lemma 5.6)

Let G be a finite abelian p-group, and let $g \in G$ be an element of largest possible order. Then

$$G = \langle g \rangle \times H \quad \text{for some subgroup } H \leq G.$$

Proof. By strong induction on $|G|$.

Base case $|G| = 1$: Then $g = 1_G$. Use $H = \{1_G\}$.

Inductive step. Assume $|G| > 1$ and the lemma holds for all abelian p-groups of smaller order. Let $|g| = p^n$ with $n \geq 1$.

Trivial case $\langle g \rangle = G$: Use $H = \{1_G\}$.

Nontrivial case $\langle g \rangle \subsetneq G$: We proceed in four steps.

Step 1: Construct $c \notin \langle g \rangle$ with $c^p \in \langle g \rangle$ and $c^p = 1$.

Take $b \in G \setminus \langle g \rangle$. Since b is a p-element, $b^{p^k} = 1 \in \langle g \rangle$ for some $k \geq 1$.

Let m be the *smallest* positive integer such that $b^{p^m} \in \langle g \rangle$ (such m exists, since $b^{p^k} = 1 \in \langle g \rangle$). Write

$$c := b^{p^{m-1}}.$$

Then:

- $c \notin \langle g \rangle$ (by minimality of m), but
- $c^p = b^{p^m} \in \langle g \rangle$, so write $c^p = g^i$ for some $i \in \mathbb{Z}$.

Claim: $p \mid i$.
Since G is a p-group with largest order p^n, $c^{p^n} = 1$, so $|c^p|$ divides p^{n-1}.

Suppose for contradiction that $\gcd(p, i) = 1$. Then by Lee Corollary 3.2 (order of a power):

$$|c^p| = |g^i| = |g| = p^n,$$

since $\gcd(i, p^n) = 1$. But $|c^p| \leq p^{n-1}$. Contradiction.

So $p \mid i$; write $i = pj$.

Construct $c' := g^{-j} \cdot c$:
Note $g^{-j} \in \langle g \rangle$. If $c' \in \langle g \rangle$, then $c = g^j c' \in \langle g \rangle$, contradicting $c \notin \langle g \rangle$.

Compute c' to the p-th power:

$$(c')^p = g^{-jp} \cdot c^p = (g^{pj})^{-1} \cdot g^i = (g^i)^{-1} \cdot g^i = 1.$$

So $|c'| = p$.

We have: $c' \notin \langle g \rangle$, $c'^p = 1$.

Step 2: Pass to $M = G/\langle c' \rangle$ and apply the inductive hypothesis.

Since G is abelian, $\langle c' \rangle \trianglelefteq G$ (automatically). The quotient $M = G/\langle c' \rangle$ is:

- abelian,

- a p-group (element orders in M divide those in G),
- of order $|G|/p < |G|$.

Claim: $|g\langle c'\rangle| = p^n$ in M.

The order of $g\langle c'\rangle$ in M must divide p^n. Suppose for contradiction that $g^{p^{n-1}} \in \langle c'\rangle$. Since $g^{p^{n-1}} \neq 1$ (as $|g| = p^n$), write $g^{p^{n-1}} = c'^s$ with $0 < s < p$.

Since $\gcd(s, p) = 1$, Bezout's theorem gives $u, v \in \mathbb{Z}$ with $su + pv = 1$. Then:

$$c' = c'^{su+pv} = (c'^s)^u (c'^p)^v = g^{p^{n-1}u} \cdot 1 \in \langle g\rangle,$$

contradicting $c' \notin \langle g\rangle$. Hence $|g\langle c'\rangle| = p^n$.

So $g\langle c'\rangle$ is an element of largest order in M. By the inductive hypothesis:

$$M = N \times K, \quad \text{where } N = \langle g\langle c'\rangle\rangle.$$

By the Correspondence Theorem (Lee, Thm. 4.8), $K = H/\langle c'\rangle$ for some subgroup $H \leq G$ containing $\langle c'\rangle$.

Step 3: Show $\langle g\rangle \cap H = \{1_G\}$.

Suppose $g^i \in \langle g\rangle \cap H$. Then $g^i\langle c'\rangle \in N \cap K$. Since $M = N \times K$ is a direct product, $N \cap K = \{e_M\}$, so:

$$g^i\langle c'\rangle = e_M = \langle c'\rangle \implies g^i \in \langle c'\rangle.$$

Since $|g\langle c'\rangle| = p^n$ in M, p^n divides i, so $g^i = 1$. Therefore $\langle g\rangle \cap H = \{1_G\}$.

Step 4: Show $G = \langle g\rangle H$.

Take any $x \in G$. In $M = N \times K$:

$$x\langle c'\rangle = (g^t\langle c'\rangle) \cdot (w\langle c'\rangle) \quad \text{for some } t \in \mathbb{Z},\ w \in H.$$

Then $x = g^t \cdot w \cdot c'^{\ell}$ for some $\ell \in \mathbb{Z}$. Since $c' \in \langle c'\rangle \subseteq H$, we have $c'^{\ell} \in H$, so $wc'^{\ell} \in H$. Hence $x \in \langle g\rangle \cdot H$.

Conclusion. $G = \langle g\rangle \times H$. ☐

Remark 1.1: The role of the largest-order hypothesis (from lecture)

The hypothesis that g has *largest* order in G is used in two critical places:

(1) **Step 1 (Claim $p \mid i$):** If $\gcd(p, i) = 1$, then $|c^p| = |g^i| = p^n$, contradicting $|c^p| \leq p^{n-1}$ (which comes from the largest order being p^n).

(2) **Step 2 (Claim $|g\langle c'\rangle| = p^n$):** If the order dropped, we would find $c' \in \langle g\rangle$, a contradiction.

Without the largest-order condition, neither step goes through. The lemma can fail for arbitrary elements.

2 The Fundamental Theorem of Finite Abelian Groups

Theorem 2.1: Fundamental Theorem of Finite Abelian Groups (from lecture; cf. Lee, Thm. 5.3)

Let G be a finite abelian group. Then G is the direct product of subgroups

$$G \;\cong\; H_1 \times H_2 \times \cdots \times H_r,$$

where each H_i is cyclic of order $p_i^{n_i}$, with p_i prime (not necessarily distinct) and $n_i \geq 1$.

Remark 2.1: Roadmap of the proof (from lecture)

The proof uses the machinery built over Lectures 10–13:

(1) **Lemma 6** (Lecture 12): $G \cong H_{p_1} \times H_{p_2} \times \cdots \times H_{p_k}$, where each H_{p_i} is the p_i-subgroup of G. Reduces to classifying p-groups.

(2) **Lemma 7** (this lecture): Any finite abelian p-group P splits as $P = \langle g \rangle \times H$, where g has largest order.

(3) **Induction on $|P|$:** Repeatedly split off cyclic factors. Each factor $\langle g \rangle$ is cyclic of prime power order p^n. The remaining subgroup H has strictly smaller order, so the inductive hypothesis applies.

(4) **Combine:** The final decomposition is a product of cyclic groups of prime power order.

Remark 2.2: Consequences (from lecture; cf. Lee, Cor. 5.1)

Equivalently, every finite abelian group is isomorphic to a direct product

$$\mathbb{Z}_{p_1^{n_1}} \times \mathbb{Z}_{p_2^{n_2}} \times \cdots \times \mathbb{Z}_{p_r^{n_r}},$$

where the p_i are (not necessarily distinct) primes. This follows from combining the Fundamental Theorem with the fact that a cyclic group of order n is isomorphic to $\mathbb{Z}_n$ (Lee, Thm. 4.14).

Example 2.1: Abelian groups of order $16 = 2^4$ (from lecture; cf. Lee, Example 5.7)

Up to isomorphism, the finite abelian groups of order 16 are exactly:

$$\mathbb{Z}_{16}, \quad \mathbb{Z}_8 \times \mathbb{Z}_2, \quad \mathbb{Z}_4 \times \mathbb{Z}_4, \quad \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2, \quad \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2.$$

These correspond to the partitions of 4 (the exponent in 2^4): 4, $3+1$, $2+2$, $2+1+1$, $1+1+1+1$.

Checklist of Skills

- ☐ State Lemma 7 and identify what "largest order" means for the induction.
- ☐ Carry out Step 1 of the proof: find $b \notin \langle g \rangle$, define $c = b^{p^{m-1}}$, show $c \notin \langle g \rangle$, show $c^p = g^i$ for some i, prove $p \mid i$ by contradiction via order estimates.
- ☐ Construct $c' = g^{-j}c$ from $i = pj$ and verify: $c' \notin \langle g \rangle$, $(c')^p = 1$.
- ☐ Carry out Step 2: pass to $M = G/\langle c' \rangle$, verify it is a smaller abelian p-group, prove $|g\langle c' \rangle| = p^n$ by showing $c' \in \langle g \rangle$ is impossible.
- ☐ Apply the inductive hypothesis to M to get $M = N \times K$, then lift to $K = H/\langle c' \rangle$.
- ☐ Carry out Steps 3 and 4: prove trivial intersection $\langle g \rangle \cap H = \{1\}$ and $G = \langle g \rangle H$ via the quotient $M = N \times K$.
- ☐ Explain why the largest-order hypothesis is essential in Steps 1 and 2.
- ☐ State the Fundamental Theorem of Finite Abelian Groups and its corollary ($G \cong \mathbb{Z}_{p_1^{n_1}} \times \cdots \times \mathbb{Z}_{p_r^{n_r}}$).
- ☐ Outline how the FTFAG follows from Lemma 6 + Lemma 7 + induction.
- ☐ List the abelian groups of order 16 and recognise they correspond to partitions of 4.